The Asymmetric Spillover Effects of Underlying Asset Volatility on NFT Liquidity

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Outline

Market Phenomenon and Initial Hypothesis

- **Market Friction:** NFT liquidity relies heavily on underlying assets (e.g., ETH). During macroeconomic turbulence (e.g., the 2022 crypto storms), NFTs frequently face severe liquidity dry-ups.

- **Initial Hypothesis (Flight to Quality):** The dynamic contagion induced by macro volatility is asymmetric. Capital panics and withdraws from tail NFT projects while concentrating on blue-chip NFTs as a relative safe haven.

- **Core Question for Defense:** Does rational “Flight to Quality” truly exist in Web3 markets lacking fundamental valuation, or is it a homogeneous liquidity collapse?

Data Engineering and Panel Construction

Processed 1.15GB of micro-level transaction data and high-frequency macro data (Jan - Jun 2022) to construct a strict Panel Data structure:

- **Objective Stratification:**
  - *Blue-chip:* Top 5% of collections ranked by historical volume.
  - *Tail:* Bottom 50% of collections.
- **Robust Treatment of Zero-trading Days:**
  - *Volume:* Handled via $\ln(1 + \text{Volume})$ to retain informative zeros.
  - *Floor Price Triple-check:* (A) Drop NaNs, (B) LOCF (Forward Fill), and (C) Missing Indicator Dummy Variable to prevent artificial liquidity inflation.

Methodology: High-Frequency Volatility Decomposition

To separate continuous volatility from discrete jumps (Barndorff-Nielsen & Shephard, 2004), we utilize high-frequency ETH data to compute:

- **Realized Volatility (RV):** Captures total price variation.

$$RV_t = \sum_{i=1}^{n} r_{t,i}^2 \tag{1}$$

- **Bipower Variation (BPV):** Robust to jumps, capturing continuous volatility.

$$BPV_t = \frac{\pi}{2} \sum_{i=2}^{n} |r_{t,i}||r_{t,i-1}| \tag{2}$$

- **Jump Component (J_t):** The discrete jump variation.

$$J_t = \max(RV_t - BPV_t, 0) \tag{3}$$

Methodology: Dynamic Identification

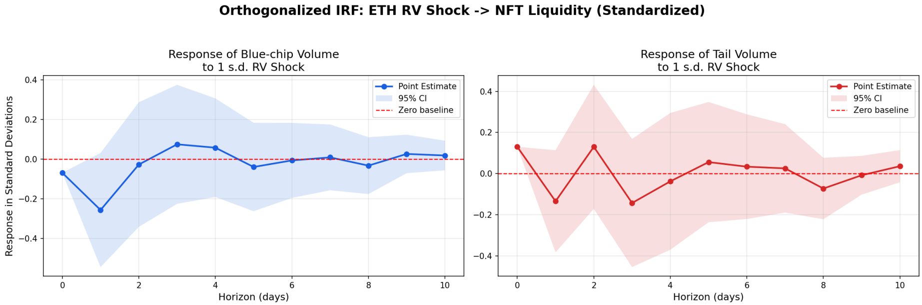
To address extreme variance explosion and endogeneity in crypto markets, we apply a dual-validation framework:

- **Stationarity & Scaling:** First-difference to extract shocks and Z-Score standardization.
- **Model 1: VAR with Economic Cholesky Ordering**
  - Causal ordering: Macro VIX $\rightarrow$ ETH RV $\rightarrow$ Blue-chip $\rightarrow$ Tail.
- **Model 2: Local Projections (Jordà, 2005)**
  - Directly regresses future NFT liquidity on current macro shocks:

$$Y_{t+h} = \alpha_h + \beta_h \text{Shock}_t + \gamma_h \text{Controls}_t + \epsilon_{t+h} \tag{4}$$

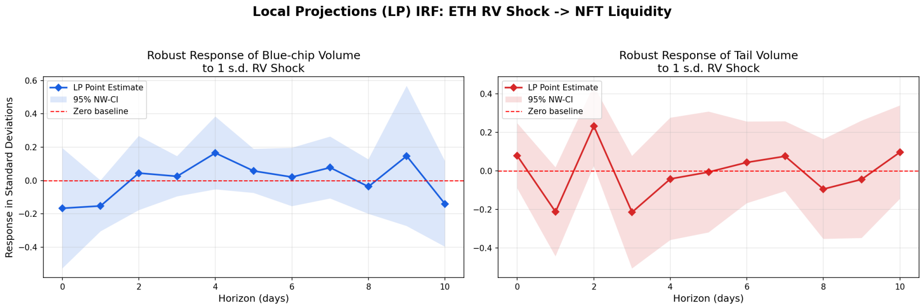
  - Incorporates Newey-West HAC standard errors for rigorous robustness checks against VAR misspecification.

Results I: Blue-chip “Liquidity Freeze”



- **Rejection of “Flight to Quality”:** Blue-chip volumes significantly **drop** at Horizon $T = 0$ to $T = 1$.
- **Diamond Hands Effect:** Holders refuse to sell at a discount, while buyers refuse to catch falling knives, leading to a complete *Liquidity Freeze*.
- *Note: Baseline IRFs display responses to total RV. Robustness checks using continuous volatility (BPV) yield consistent microstructural freezing patterns.*

Results II: Tail Asset “Panic Selling”



- **Robust Validation (LP Method):** Tail assets exhibit severe volume spikes and extreme oscillation immediately following the shock.
- **Speculative Stomping:** Driven by panic selling from weak consensus and subsequent speculative trades (Degens) betting on rebounds, revealing extreme *Fragility*.

Conclusion and Contributions

- **Core Finding: Liquidity Stratification**
  - During extreme crypto storms, there is no safe haven. Underlying volatility contagion is highly asymmetric: Blue-chips suffer from *illiquidity (freezing)*, whereas Tail assets suffer from *excessive speculative volatility (stomping)*.
- **Academic Contributions**
  - Successfully challenges the assumption of homogeneous risk exposures in digital collectibles.
  - Introduces Local Projections (LP) with micro-level transaction data to the study of Web3 market microstructure.

Thank You!